

Predictive Event-Driven Traffic Intelligence

Theme Selected: Theme 2 – Event-Driven Congestion (Planned & Unplanned) **Prepared By:** Ved Naman

1. Executive Summary

The Bengaluru Traffic Police (BTP) currently relies on reactive, experience-driven resource deployment to manage traffic breakdowns caused by events (protests, accidents, VIP movements). To solve this operational challenge, we engineered the **BTP AI Command Center**: a full-stack, predictive intelligence platform. By mathematically fusing raw spatial traffic demand data with historical BTP event logs, we trained a highly resilient Machine Learning ensemble. This model is served via a live Python FastAPI backend to a front-end dashboard, allowing officers to input upcoming events and receive real-time, actionable deployment strategies.

2. The Data Engineering Architecture (The "Spatial Bridge")

The core challenge of this problem statement was dataset asymmetry: the Flipkart traffic demand dataset utilizes MapMyIndia **geohash** grids, while the BTP event dataset utilizes exact **latitude** and **longitude** coordinates.

To bridge this gap, we implemented a **Spatial Bridge Pipeline**:

- **Geohash Encoding:** Utilized the **pygeohash** library to convert thousands of BTP GPS coordinates into Level-6 geohashes, mathematically aligning the two datasets.
- **Event Severity Scaling:** Mapped categorical event causes into a numerical "Ripple Effect" severity scale (e.g., Vehicle Breakdowns = Level 1; Massive Protests/VIP Movements = Level 5).
- **V8 Feature Engineering:** Engineered complex features to capture traffic flow, including spatial clustering (**geo_zone**, **geo_district**), environmental states (**RoadType_Weather**), and cyclic temporal transformations using sine/cosine functions for hours and minutes.

3. The Machine Learning Engine (The "Holy Trinity" Ensemble)

To ensure maximum predictive accuracy, we abandoned single-model architecture in favor of a robust gradient boosting ensemble we call the "Holy Trinity":

- **CatBoost:** Chosen for its native, highly efficient handling of categorical strings (geohashes).
- **LightGBM:** Chosen for its extremely fast, leaf-wise tree growth, ideal for large tabular datasets.
- **XGBoost:** Chosen for its historical reliability and aggressive penalization of over-fitting.

The models were trained on 100% of the newly fused dataset, learning not just baseline traffic demand, but exactly how a Level 5 event impacts specific geohashes dynamically.

4. Enterprise-Grade System Resilience (The Fallback Mechanism)

A critical requirement for law enforcement software is 100% uptime. During API development, we identified that strict tree-based models (like LightGBM) can fail during live single-row inference if the inference dictionary does not perfectly match the training categorical dictionary.

Instead of allowing the server to crash, we engineered an **Enterprise Fallback Mechanism**. If a localized data discrepancy causes one model to panic, the FastAPI server catches the error, safely bypasses the failing model, and dynamically averages the predictions of the surviving models (e.g., CatBoost). This guarantees the BTP dashboard will *always* generate a deployment plan, regardless of edge-case data inputs.

5. The Product: BTP Command Center Dashboard

We elevated the solution from a static Jupyter Notebook to a production-ready software application.

- **The Backend:** A Python FastAPI server hosting the exported `.pkl` models, featuring CORS middleware to communicate with web clients.
- **The Frontend:** A sleek, dark-mode HTML/CSS/JS interface designed for police dispatchers.
- **Business Logic Integration:** The API translates raw mathematical predictions (0.0 to 1.0+) into actionable business logic. A low score recommends a "Routine Patrol," while a score above 0.7 triggers the UI to turn red with the alert: *"URGENT: Severe Gridlock Expected! Deploy barricades and divert traffic."*

6. Conclusion and Impact

This solution directly answers the Theme 2 problem statement. We have successfully quantified the impact of unplanned and planned events in advance, removing the guesswork from police deployment. By combining advanced spatial data engineering with a resilient, full-stack software architecture, the BTP Command Center serves as a scalable, highly practical prototype for modernizing traffic law enforcement in Bengaluru.

